

Impact of Policymaking on UPI Growth in India

Statistical Analysis of Digital Payments

Subject: Statistical Methods

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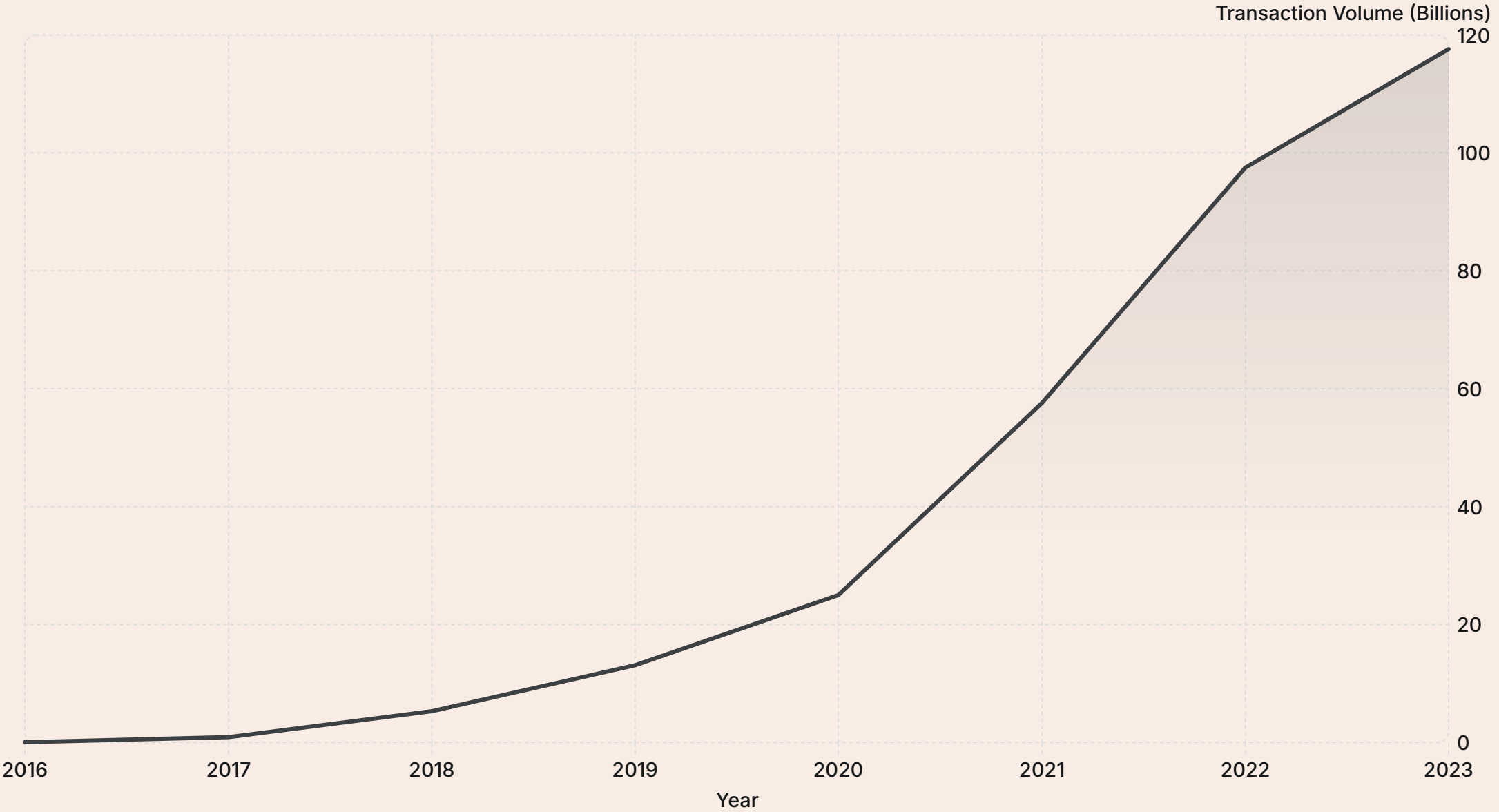


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Introduction to UPI: Revolutionizing Indian Payments

UPI is India’s real-time payment system developed by NPCI. It enables instant bank-to-bank transfers through a simple digital identifier.

UPI supports India’s shift toward a cashless economy and expands financial inclusion through fast, easy digital transactions.



The chart shows UPI’s rapid growth from 2016 to 2023, surpassing 100 billion transactions in 2023.



Problem Statement

This analysis examines whether key policies and events drove UPI's rapid growth in India.

Key Influences

- **Demonetization (2016)**
Boosted digital payment adoption.
- **COVID-19 Lockdown (2020)**
Accelerated contactless transactions.
- **Zero MDR Policy (2020)**
Improved merchant acceptance.
- **UPI Lite & Credit on UPI (2022)**
Expanded access and convenience.

Why It Matters

Digital Economy

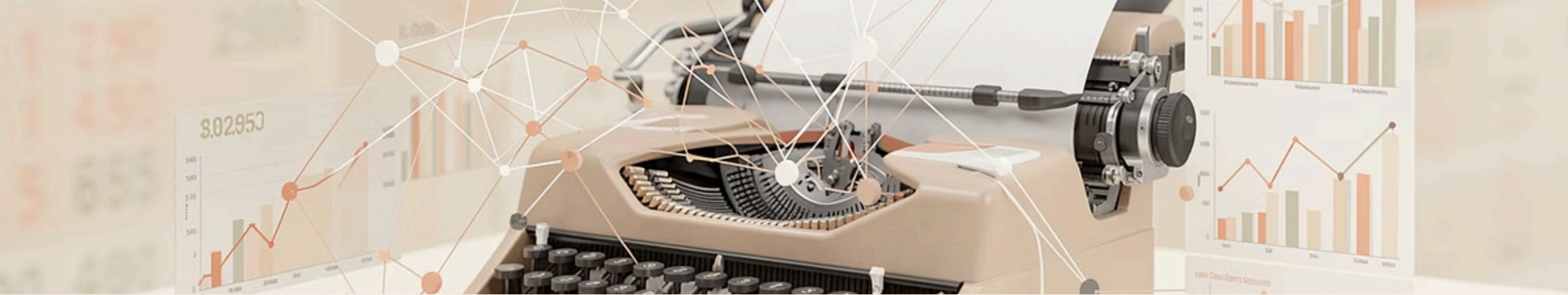
Supports broader payment adoption.

Financial Inclusion

Extends access to more users.

Cashless Growth

Strengthens a transparent payments system.



Dataset & Variables

We use a monthly time-series dataset to analyze UPI growth and policy impact.

- **Source:** RBI & NPCI
- **Period:** Jan 2016 - Dec 2026
- **Observations:** 120 monthly points

Key variables:

Variable	Description
Transaction Volume	Monthly UPI transactions (billions)
Transaction Value	Monthly UPI value (INR crores)
Avg Ticket Size	Value per transaction
Policy Variables	Indicators for major policy and event dates

Objectives & Hypothesis

This study examines the main drivers of UPI growth and tests whether policy interventions changed its growth trajectory after 2020.

Key Objectives

01

Analyze Growth

Identify major UPI growth phases and patterns.

02

Assess Policy Impact

Measure the effect of key events on UPI adoption.

03

Test Relationships

Quantify links between interventions and transaction volume/value.

04

Check Post-2020 Shift

Determine whether growth changed significantly after 2020.

Hypothesis

H₀

No significant difference in UPI growth before and after 2020.

H₁

A significant difference exists before and after 2020.

The statistical test will show whether to reject the null hypothesis.

Data Cleaning & Feature Engineering

The raw data is cleaned and transformed into a structured dataset for analysis.

Key Steps:

Date Standardization
Align dates to one consistent format.

Missing Values
Impute or remove gaps to preserve data quality.

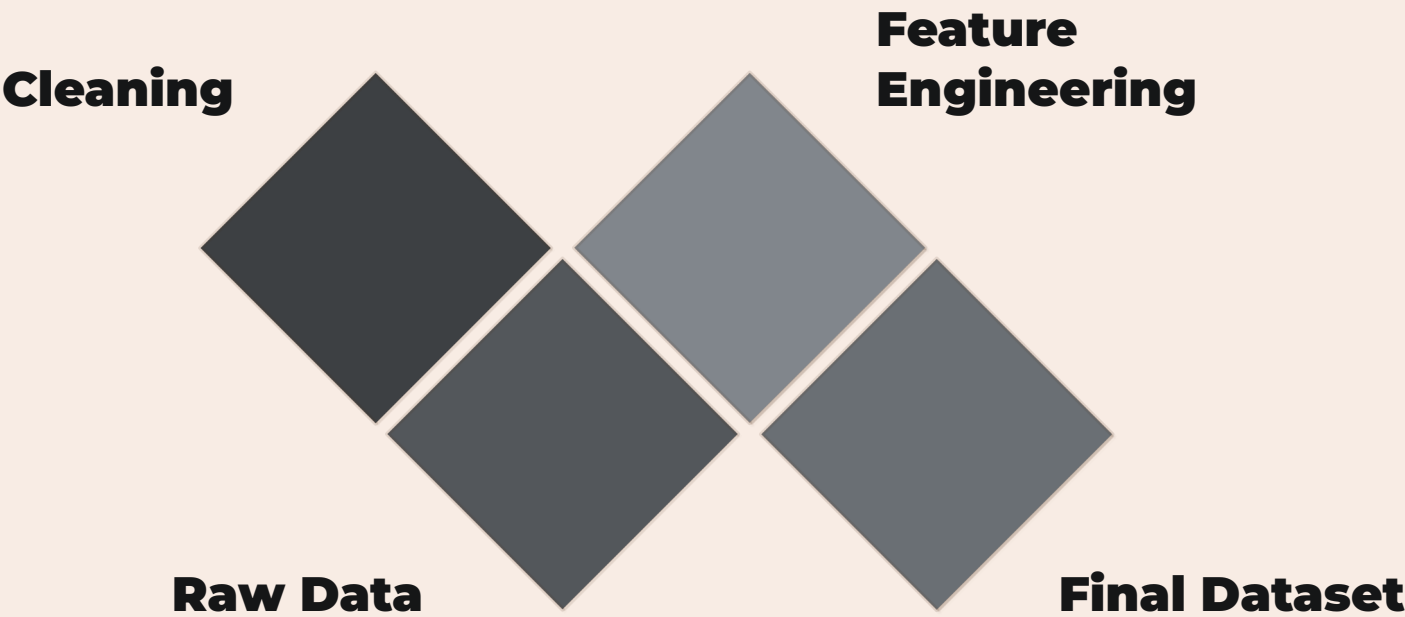
Unit Conversion
Convert values to consistent units for comparison.

Feature Engineering
Create variables that improve model performance.

Derived Variables:

Variable	Purpose
Avg Ticket Size	Value per transaction.
MoM Growth %	Monthly growth in volume and value.
Policy Flags	Binary indicators for key events.

Workflow:



These steps produce a clean, analysis-ready dataset for statistical testing.



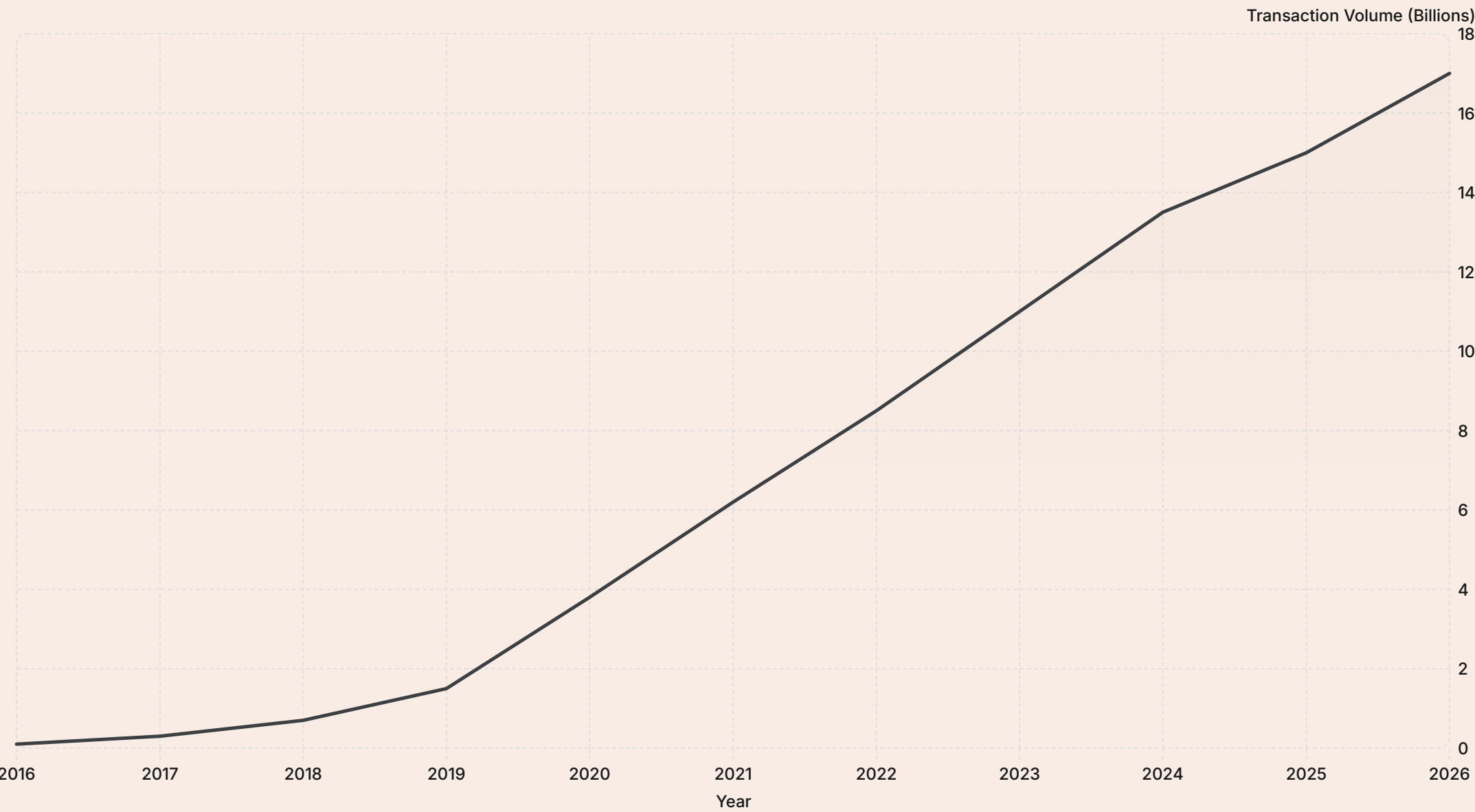
Exploratory Data Analysis (EDA)

UPI adoption shows strong growth, with clear acceleration after 2020 and visible policy-driven inflection points.

KEY OBSERVATIONS

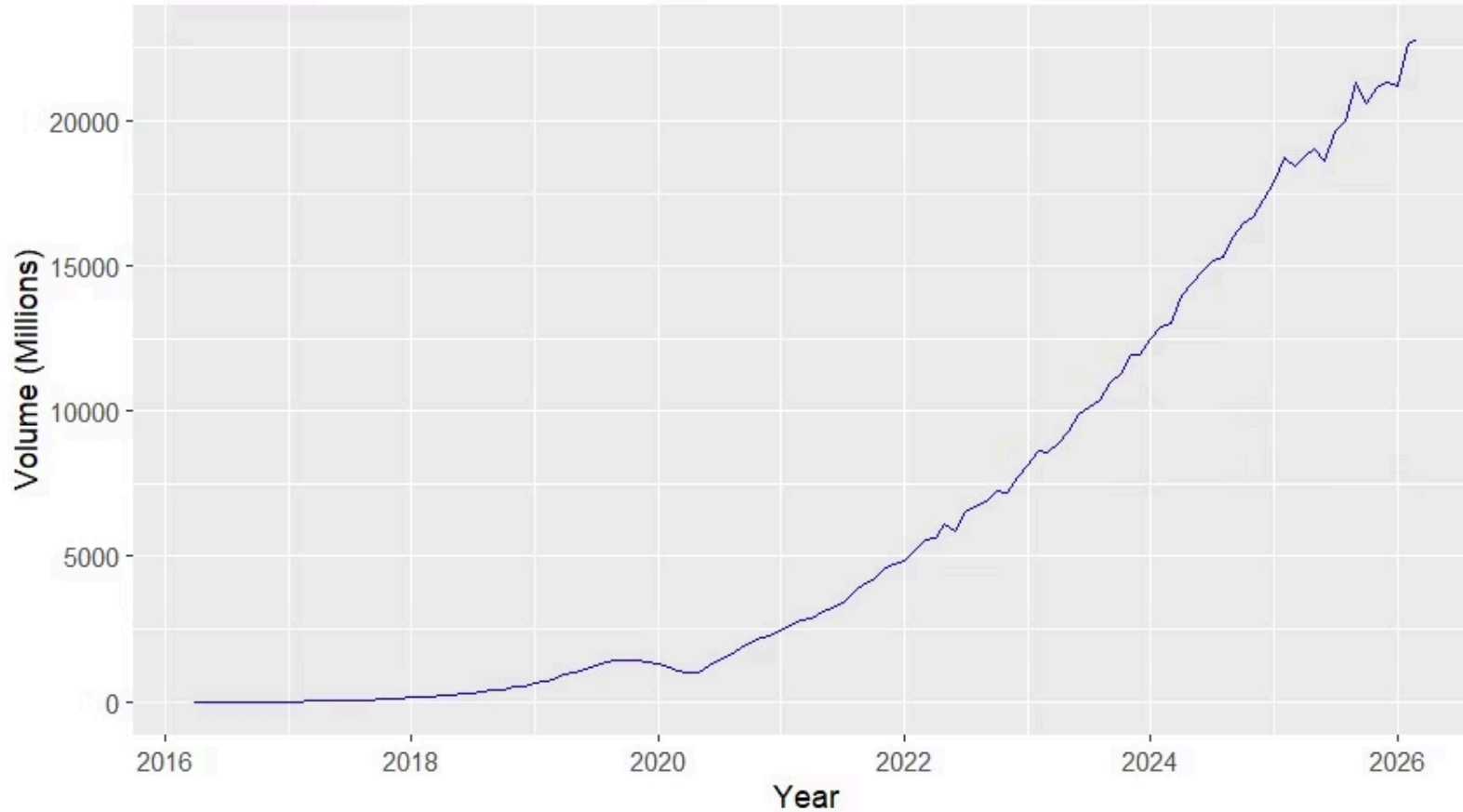
- Exponential Growth**
Transaction volume and value rise steadily over time.
- Post-2020 Acceleration**
Growth speeds up sharply after 2020.
- Policy Impact**
Policy changes appear to have catalyzed adoption.

UPI Growth Over Time



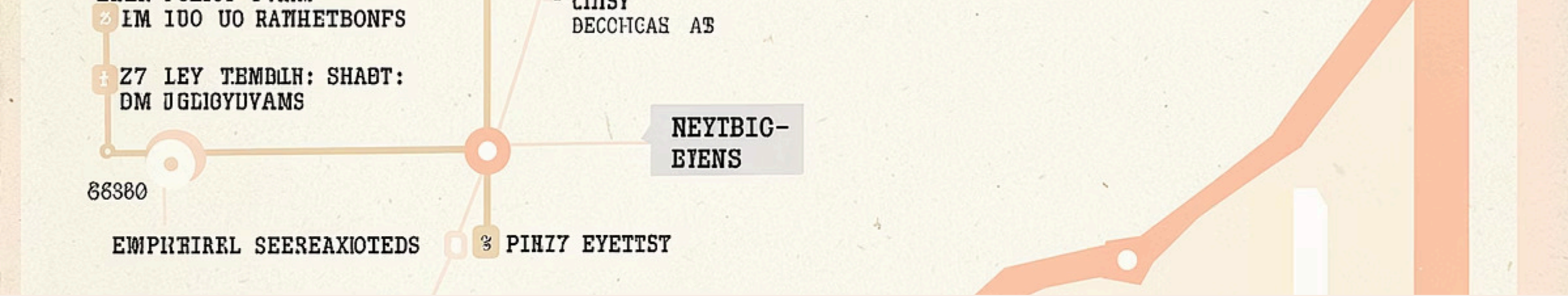
Growth is slow from 2016-2019, then accelerates sharply after 2020.

UPI Transaction Volume Growth Over Time



Interpretation:

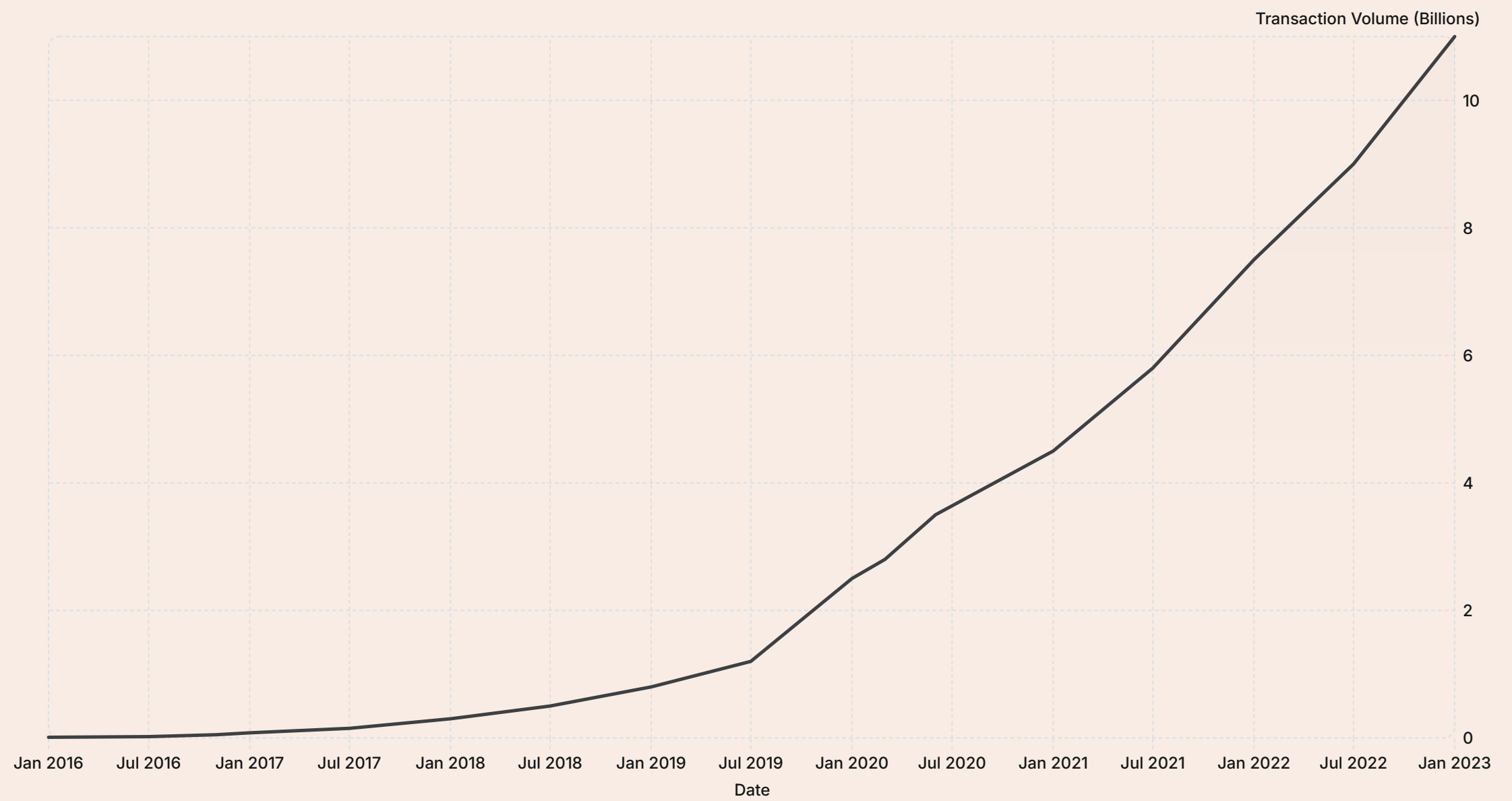
The graph shows exponential growth in UPI transaction volume over time. Growth was gradual during the early years but accelerated rapidly after 2020, indicating increasing adoption of digital payments and the expansion of India's cashless economy.



Policy Event Analysis

Policy interventions acted as key inflection points, accelerating UPI adoption.

EVENT-DRIVEN GROWTH

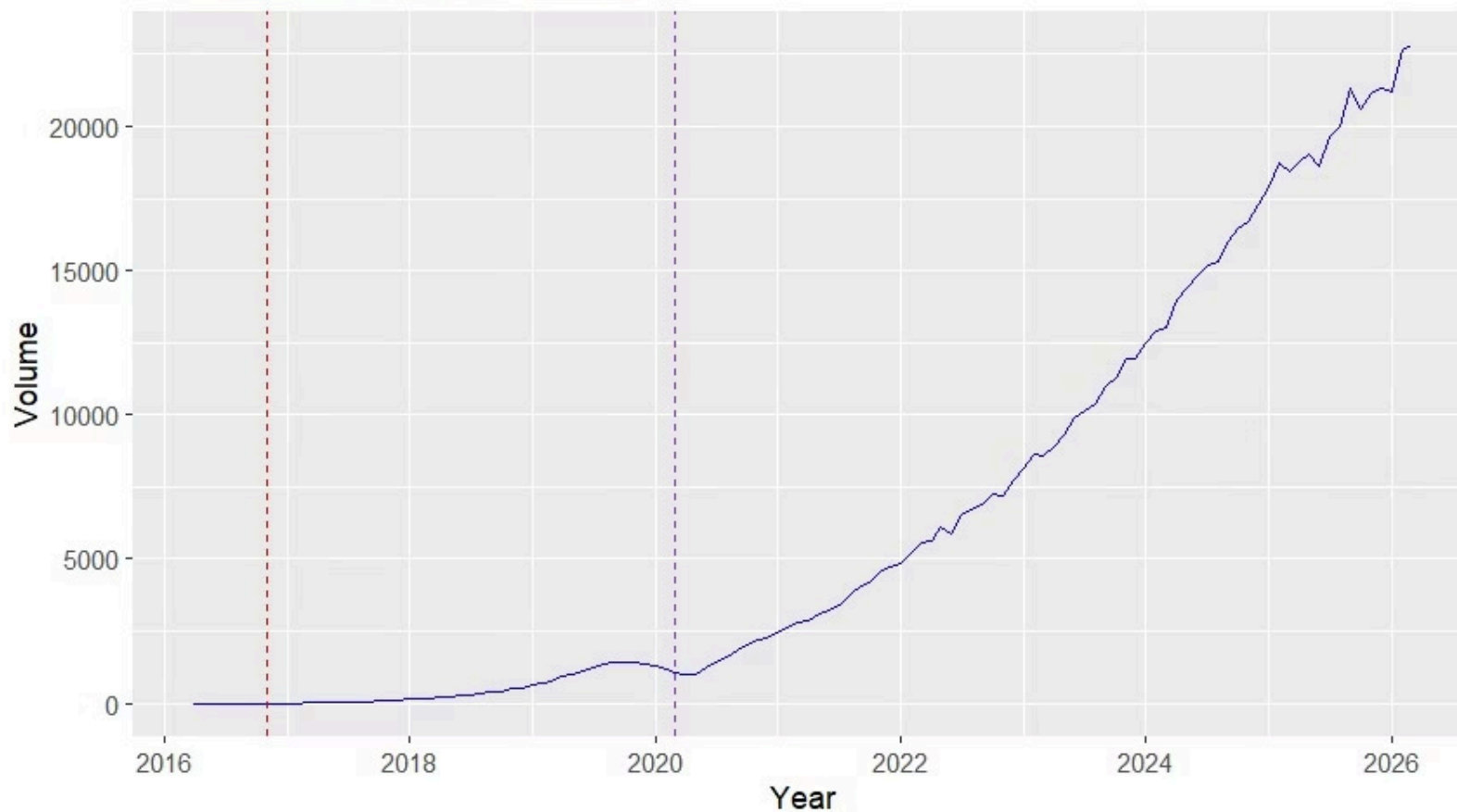


Key events highlighted in the chart:

- **November 2016:** Demonetization boosted digital payment adoption.
- **March 2020:** COVID-19 lockdown accelerated contactless payments.

Both events coincide with clear jumps in UPI usage.

UPI Growth with Policy Events



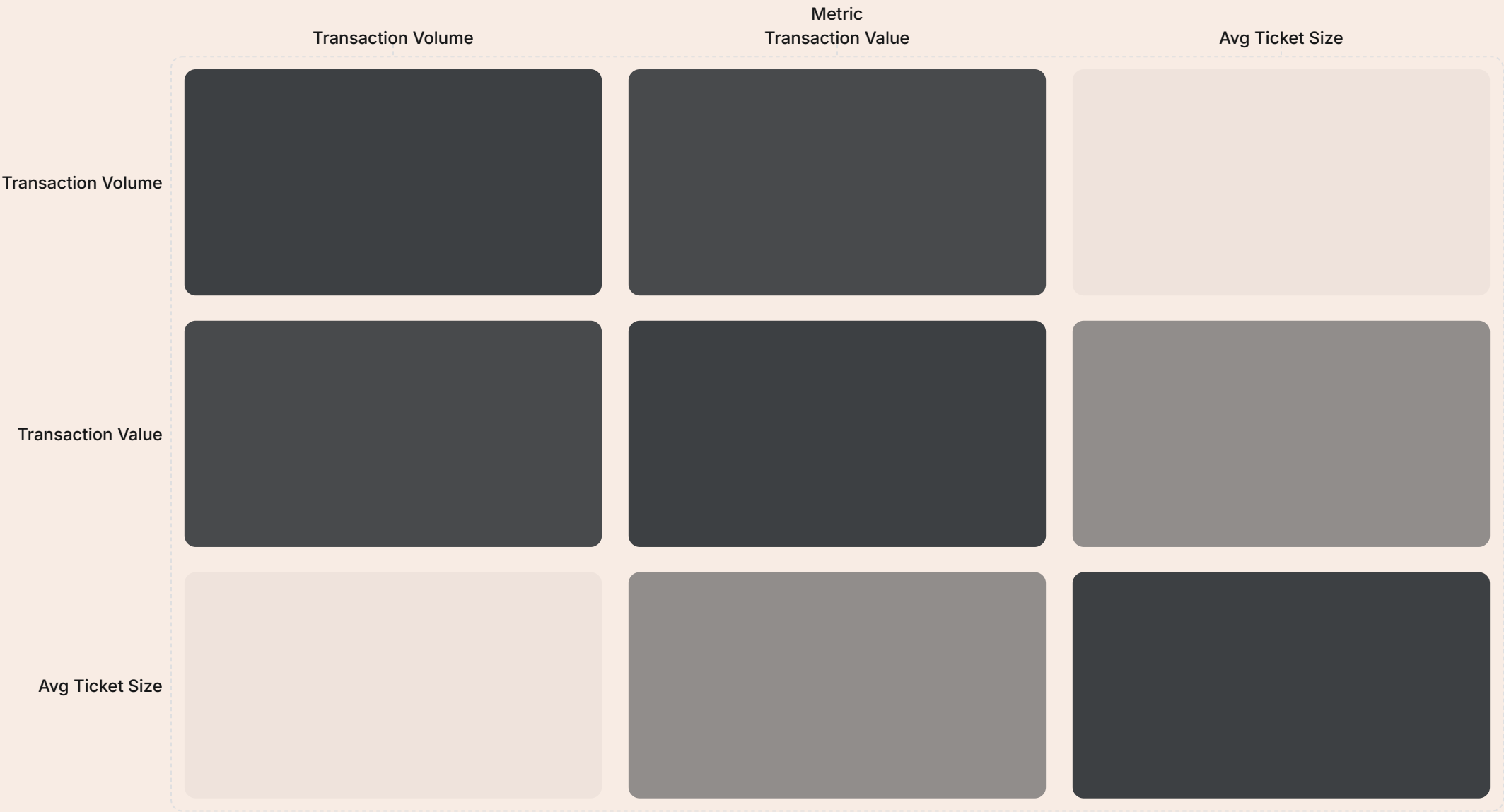
Interpretation:

UPI transaction volume shows rapid long-term growth from 2016-2026. Demonetization (2016) initiated digital payment adoption, while the COVID-19 period (2020) accelerated contactless transactions, leading to sharp growth in UPI usage after 2020.

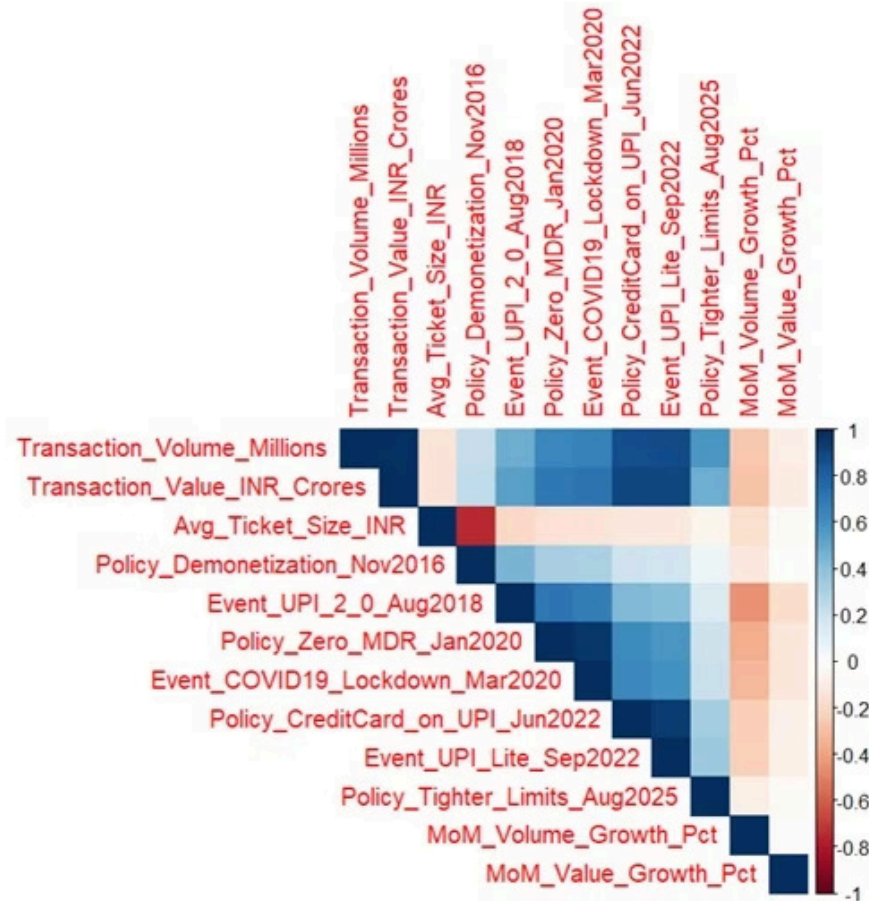
Correlation Analysis

This analysis highlights the key relationships between UPI metrics and what drives its growth.

KEY RELATIONSHIPS



The strongest link is between Transaction Volume and Transaction Value (0.95). Avg Ticket Size is only weakly related to volume (0.15), showing UPI growth is mainly **volume-driven**.



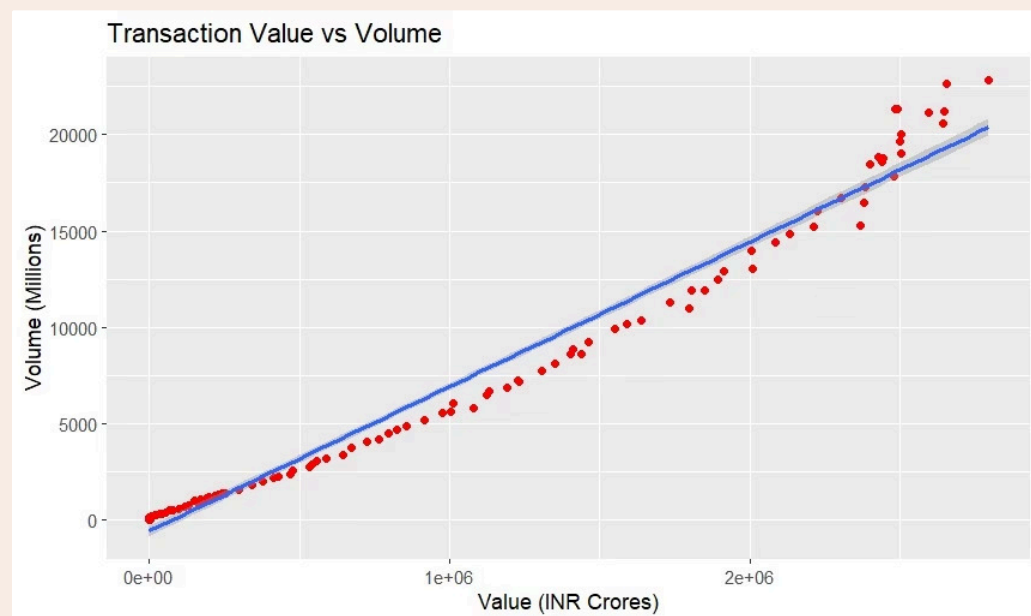
Interpretation:

The correlation heatmap indicates a strong positive relationship between UPI transaction volume and transaction value, showing that both grew simultaneously over time. It also suggests that policy interventions had a meaningful impact on accelerating digital payment adoption in India.

Scatterplot Analysis

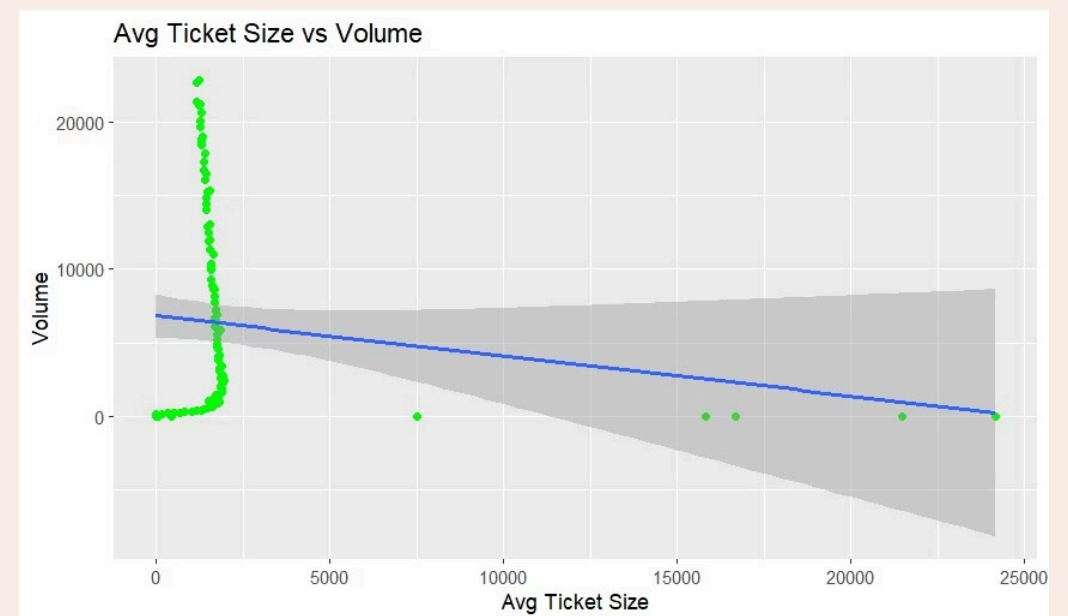
Our scatterplot analysis highlights the key relationships driving UPI's growth.

Transaction Value vs. Volume



Strong positive relationship.

Avg Ticket Size vs. Volume



Weak relationship; ticket size stays relatively stable.

Growth is driven by more transactions, not larger transactions.

UPI's success comes from high-frequency, smaller-value payments.



Outlier & Distribution Analysis

UPI transaction data is right-skewed and includes outliers, making robust analysis essential.

KEY INSIGHTS FROM DATA DISTRIBUTION

Right-Skewed

Most transactions are small, with fewer large values.

Outliers

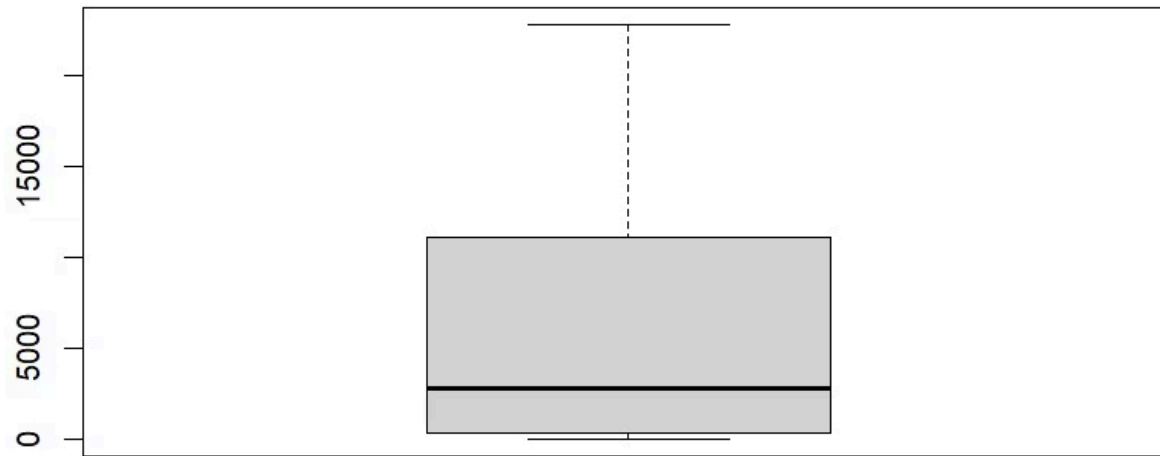
Early adoption spikes and policy events create extreme values.

Non-Normal Data

Use methods suited to skewed financial data.

i **Non-parametric tests** will reduce distortion from outliers and distribution assumptions.

UPI Volume Outliers



Interpretation:

The correlation heatmap indicates a strong positive relationship between UPI transaction volume and transaction value, showing that both grew simultaneously over time. It also suggests that policy interventions had a meaningful impact on accelerating digital payment adoption in India.



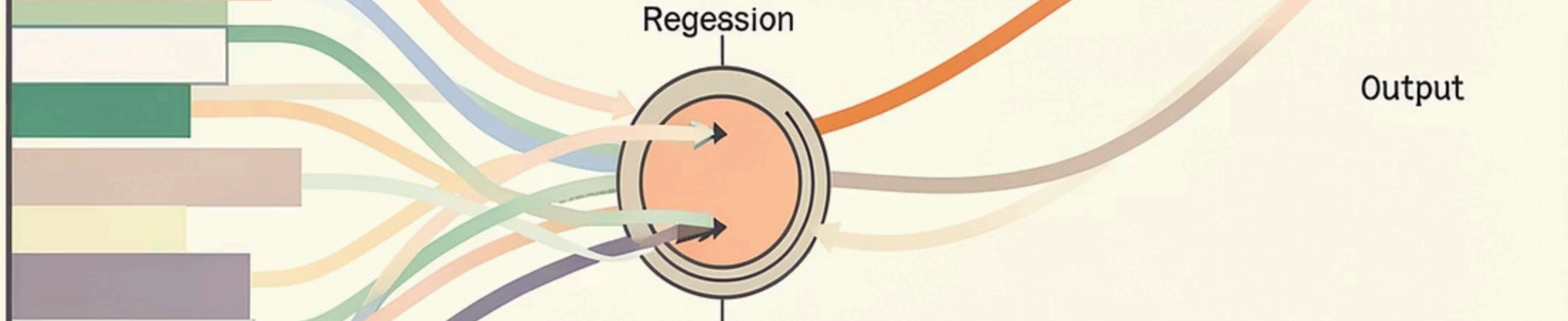
Statistical Methodology

We use a focused set of statistical techniques to measure policy impact on UPI growth.

ROBUST ANALYTICAL APPROACH

Method	Purpose
Regression Analysis	Measure direct policy effects.
Wilcoxon Signed-Rank Test	Compare growth across policy phases.
Correlation Analysis	Identify key variable relationships.
VIF Analysis	Check multicollinearity.
Moving Averages	Reveal trends in noisy time-series data.

These methods provide a reliable basis for interpreting UPI's evolution.



Multiple Linear Regression

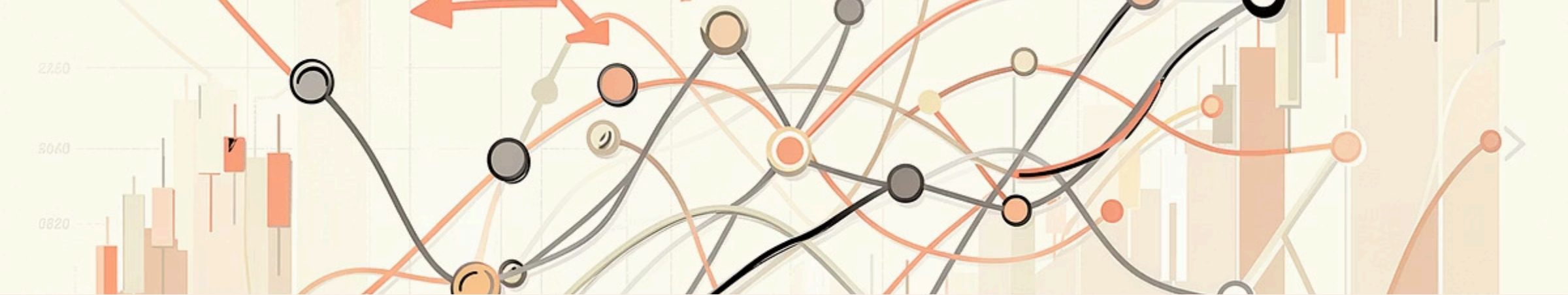
We use Multiple Linear Regression to measure how policy interventions and other variables affect UPI transaction volume.

QUANTIFYING POLICY IMPACT

$$Y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \cdots + \epsilon$$

Metric	Value
R ²	0.872
Adjusted R ²	0.865
F-statistic	128.3
p-value	<0.001

The model explains 87.2% of the variation in UPI transaction volume.



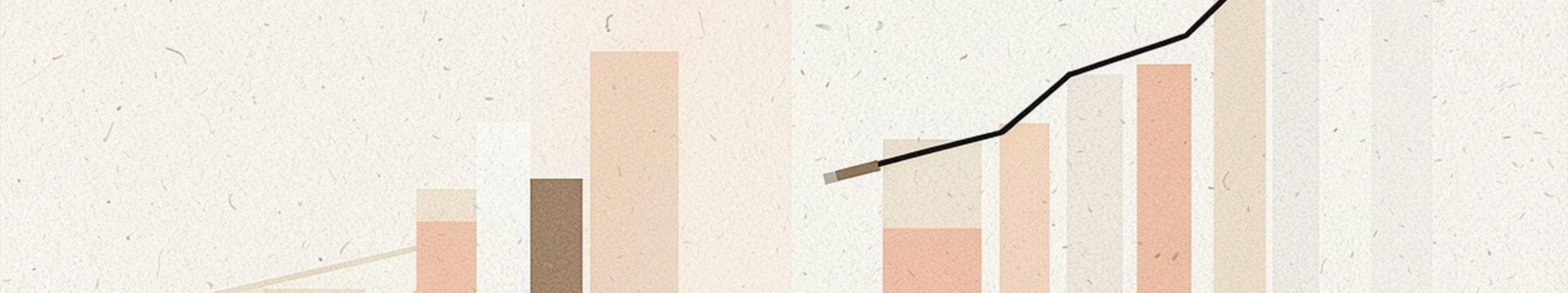
Significant Policy Variables

Key interventions and external factors shaped UPI's growth.

KEY POLICY DRIVERS

Variable	Significance
Credit Card on UPI	Significant
Tighter Limits	Significant
COVID-19 Pandemic	Not significant

Credit card integration and tighter limits were the strongest drivers. COVID-19 boosted adoption, but was not independently significant in the model.



Wilcoxon Test Results

We used the non-parametric Wilcoxon Signed-Rank Test to assess UPI growth before and after the 2020 policy shift.

WHY THE WILCOXON TEST?

Handles Skewed Data

UPI transaction data is highly **right-skewed**, making Wilcoxon well suited for this analysis.

No Normality Assumption

Unlike t-tests, the Wilcoxon test does **not require normality**, making it more robust here.

Key results: $W = 2593$ and $p = 2.413 \times 10^{-6}$.

Reject H_0 : Significant Difference in Growth Trajectory

The very low p-value confirms a statistically significant change in UPI growth after 2020.



Key Findings & Policy Implications

SYNTHESIS OF INSIGHTS

Our analysis highlights the main drivers behind UPI's rapid growth and its impact on India's digital economy.

Key Findings

1	Policy Support Government backing and regulation fueled UPI's growth.
2	COVID-19 Boost The pandemic accelerated digital payment adoption.
3	Everyday Payments High volumes of small transactions drive UPI's use.
4	New Integrations Credit card support can further expand UPI usage.
5	Digital Shift These trends reflect India's move toward a digital economy.

Real-World Implications

- **Financial inclusion:** More people can access formal finance.
- **Less cash dependence:** Payments become faster and more secure.
- **Greater transparency:** Transactions are easier to track.



Limitations and Future Scope

ACKNOWLEDGING ANALYTICAL BOUNDARIES

Our analysis offers useful insights, but key limitations shape how the results should be interpreted.

Limited Variables

We focused on direct policy effects, leaving out some broader macroeconomic factors.

Policy Overlap

Closely timed interventions made it difficult to isolate individual impacts.

Time Dependence

Past trends strongly influence future UPI patterns, complicating causal analysis.

Future Uncertainty

Policy shifts or economic shocks could change adoption trajectories.



Conclusion

KEY TAKEAWAYS

UPI's growth highlights how policy has accelerated digital payment adoption in India.

Strong Growth

UPI expanded rapidly and became central to India's payment ecosystem.

Policy Matters

Government support played a major role in adoption and usage.

Structural Shift

Post-2020, payment behavior changed faster and more permanently.

Digital Foundation

UPI is now a core pillar of India's digital economy.

“UPI reflects India's transition to a policy-driven digital economy.”

References

Our analysis draws on authoritative sources to ensure accuracy.

- **Regulatory bodies:** RBI publications and policy documents.
- **Payment ecosystem:** NPCI reports, circulars, and data releases.
- **Government sources:** Official reports, economic surveys, and policy papers.
- **Research and media:** Peer-reviewed studies, industry reports, and reputable news coverage.